



Semantic Research Paper Search Engine

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Motivation and Review

Traditional keyword search fails to capture semantic similarity between documents and queries. Researchers often miss relevant papers because keyword matching cannot understand meaning.

Recent advances in sentence embedding models (e.g., models from Hugging Face) allow text to be represented as dense vectors, enabling semantic search.

This project builds a semantic research paper search engine that retrieves papers based on meaning rather than exact keywords. The system will enforce data contracts to ensure dataset quality and consistency, and will be deployed with Docker and an interactive Streamlit interface.

Datasets

- ArXiv dataset
- Semantic Scholar dataset
- CORD-19 dataset

Methodology

- Collect research paper data containing title, abstract, authors, publication year, and paper ID from publicly available datasets.
- Define a data contract schema specifying required fields and data types.
- Embedding Generation: Generate dense vector representations using pretrained sentence embedding models from Hugging Face such as all-MiniLM-L6-v2.
- Semantic Retrieval: Store embeddings in a vector index (e.g., FAISS) and perform similarity search to retrieve the top-k semantically similar documents for a given query.
- Interface & Deployment: Build an interactive search interface using Streamlit and deploy the entire pipeline in a reproducible environment using Docker.

Novelty

The novelty of this work lies in integrating semantic search with ML-Ops principles. Unlike typical embedding-based retrieval systems, this project introduces data contracts for dataset validation, metadata-aware ranking for contextual retrieval, and embedding drift monitoring. The system is implemented as a fully containerized ML pipeline with an interactive search interface, demonstrating a production-ready semantic retrieval framework.

Expected Outcomes

The goals of this project are:

- Retrieve semantically similar research papers.
- Improve search relevance over keyword search.
- Provide a reproducible ML pipeline with data validation.
- Demonstrate a deployable ML application.

Metrics

I would like to extensively test my approach across a comprehensive suite of metrics

- Precision@k
- Recall@k
- nDCG (Normalized Discounted Cumulative Gain)
- query latency
- throughput